

JEDIDIAH ROSEMAN

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EDUCATION

University of Michigan

Bachelor of Science in Electrical Engineering | Major GPA: 3.91

Ann Arbor, Michigan

Expected May 2028

Relevant Coursework: EE System Design, Intro Circuits, Semiconductor Devices, Signals & Systems, Electromagnetics, Programming & Data Structures, Probabilistic Methods in Engineering, Multivariable Calculus, Differential Equations

TECHNICAL PROJECTS

Autonomous Warehouse Inventory Robot

Independent Hardware Project | ROS2, Raspberry Pi, Arduino, RPLIDAR

Atlanta, Georgia

Summer 2026 - Present

- Building a robot to count \$4.4M of inventory across 1,598 SKUs at Payless Decor, replacing a two-person manual count that takes 7-14 workdays on ladders and forklifts; designed to run overnight so counts can go from quarterly to monthly without pulling SKUs off sale.
- Caught that the vendor encoder spec assumed 4× quadrature decoding, which the ATmega328P's two hardware interrupts could not support; redesigned for 2× decoding and corrected the odometry constant to 0.62 mm/count before navigation tuning.
- Sized the power system around a 12.8 A worst-case four-motor stall, cutting the main fuse rating from 40 A to 20-25 A so protection trips ahead of the BMS threshold and a fault clears with a two-dollar part instead of a manual battery pack reset.
- Wrote the Raspberry Pi-to-Arduino serial firmware with framed motor commands, a 20 Hz odometry stream, and a 500 ms watchdog that halts drive on link loss; diagnosed and cleared 5 hardware and spec defects during bring-up, including misaligned motor inserts.

Sensor-Based Attendance Counter

ENGR 100 Engineering Capstone, University of Michigan

Ann Arbor, Michigan

Spring 2025

- Built a real-time bidirectional people counter on a 5-person team using dual ultrasonic sensors (chosen over infrared and camera-based options) and an Arduino in a 3D-printed door-frame enclosure modeled in Onshape, detecting entry vs. exit by order of occlusion.
- Achieved 98% counting accuracy across repeated lecture hall trials on a self-contained battery-powered unit built for under \$40 in parts; owned all hardware integration including sensor wiring, power routing, and characterization across distance and angle.

Midtown Electric Vehicle Rally

Co-Founder and Head of Logistics

Atlanta, Georgia

2022 - 2024

- Engineered the full wiring system for a brushless DC motor drivetrain, collaborating on electrical and mechanical integration across a 20+ member team to field a 65 MPH vehicle that placed in national 2-hour endurance races across two competition seasons.
- Secured \$25K in sponsorship from Mercedes-Benz and Rivian via executive outreach to launch Atlanta's first high school EV club.

LEADERSHIP

Michigan Defense Technology (MDT)

Co-President and Founder

Ann Arbor, Michigan

2025 - Present

- Led outreach and technical scoping for 3 industry-partnered projects, turning sponsor requirements into deliverables for 6-10 person student teams: a drone swarm kit for AG3, an AUV research platform with Booz Allen, and a wind tunnel data pipeline for North Wind.
- Co-founded and scaled U-M's first defense tech organization to 100+ applicants at an 18% acceptance rate while defining the strategic roadmap for technical research, operations, recruitment, and partnerships with Palantir, Anduril, Scale AI, Shield AI, and Antares.

WORK EXPERIENCE

First Financial Bank USA

Product Manager, Contracted Software Project

Atlanta, Georgia

Summer 2026

- Directed a 7-member cross-functional team in a 13-week sprint to deliver a full-stack fintech MVP targeting \$50M in retail IRA deposits, clearing all 3 executive gate reviews on schedule while coordinating workstreams across 3 structured development phases.
- Architected an end-to-end product build spanning frontend engineering, backend systems, quantitative modeling, and compliance, delivering a production-ready prototype with full technical documentation and handoff materials for external integration.

Additional Experience: Investment Analyst, Service Provider Capital (2025); VP of Consulting, Ascend Consulting Group (2024 - Present)

SKILLS & HONORS

Hardware: Circuit design, PCB schematic capture and layout, soldering and crimping, brushless DC motor drivetrains, hardware bring-up and fault isolation, sensor integration (LIDAR, ultrasonic, IMU, camera), UART and I2C, oscilloscope, multimeter, 3D printing

Software & Tools: C++, Python, JavaScript, TypeScript, React, HTML, MATLAB, LTspice, Altium, KiCad, ROS2, Arduino, Raspberry Pi, Onshape (CAD), Git, Linux, Docker, Postgres, Vercel, Excel, STAR.SDK, Cursor, Notion

Honors: Harvard National Debate Tournament Quarter Finalist (Top 100 in US); MIT Inspirit AI, built a CNN chest X-ray pneumonia classifier

Hobbies: App Development, Video Production, Backpacking, Basketball, Golf, Music (Vinyl)